Lab 4 - Autonomous Line Tracking

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[1] Data Processing & Experiment Implementation[1]

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Abstract

This experiment focuses on implementing line-tracking behavior for a robotic car, with the primary objective of enabling it to perform precise path-following tasks based on infrared (IR) sensor feedback. Specifically, the study investigates the variation in analog-to-digital converter (ADC) values of incoming IR signals as the car moves at different distances from the guiding line. By analyzing these signal changes, we aim to establish a quantitative relationship between the IR sensor outputs and the car's lateral position relative to the path. Furthermore, the experiment explores how different line shapes affect the IR sensor's performance, thereby providing insight into how environmental factors influence the accuracy and stability of line tracking. The results offer foundational data to support autonomous navigation of the car.

1 Introduction

1.1 Experiment 1

Investigating the Relationship Between the Distance of the TCRT5000 Sensor from the Ground and the Intensity of Reflected Infrared Light

We investigated the variation of infrared light reflection intensity with distance between 5 mm and 40 mm and plotted the results into a graph.

1.2 Experiment 2

Testing whether the three sensors can correctly identify black lines and examining their recognition performance at different positions of the black lines

We tested the sensor's recognition performance under seven different states, which are represented as 000, 001, 011, 010, 100, 110, and 111 (assuming that detecting white is represented by 1 and detecting black by 0).

2 Theory and Technical Background

The TCRT5000 1 is a small photoelectric component that belongs to the category of infrared invisible light products. It is capable of detecting changes in light intensity. The infrared-emitting diode of this device continuously emits infrared light. When the emitted infrared light is not reflected or the reflected intensity is insufficient, the phototransistor remains in the

cut-off state. The module's output terminal is at a low voltage level, and the indicator diode stays off. When an object to be detected enters the detection range, the infrared light is reflected back with sufficient intensity. The phototransistor then enters the saturation state, the module's output terminal switches to a high voltage level, and the indicator diode is illuminated.



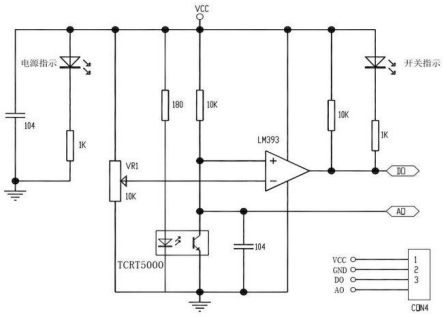


Figure 1: TCRT5000 Components and Schematic Diagrams

3 Method

Abridge	Full Name
ADC	Analog-to-Digital Conversion
IDE	Integrated Development Environment
IR	Infrared Transducer

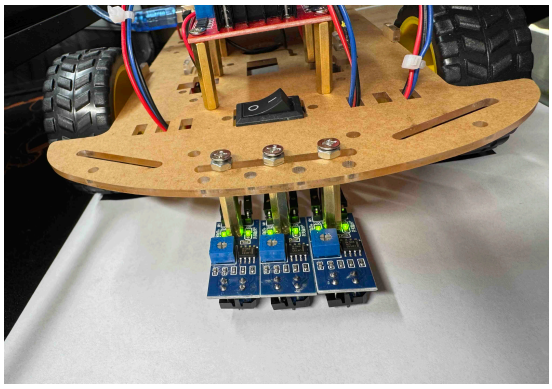
Keywords: IDE, IR, Variable Control

3.1 Experiment 1

We measured the changes in the ADC values of the IR sensor at intervals of every 0.5 cm from a white sheet of paper, obtaining a total of 10 sets of data ranging from 0.5 cm to 5 cm. The program is written using Arduino IDE to obtain the voltage value from analog pin A0 and convert it to a digital value through ADC. The corresponding sensorValue is obtained through the Serial Monitor in the IDE.

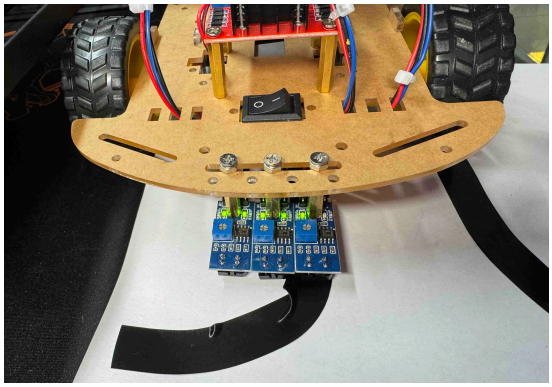
3.2 Experiment 2

In this experiment, the sensors were installed at a height of 80 millimeters above the chassis. The program is written using Arduino IDE to obtain the voltage value from analog pin A0 and convert it to a digital value through ADC. The corresponding sensorValue is obtained through the Serial Monitor in the IDE. By reading the sensor values, we can accurately determine whether each sensor is positioned over a white surface or a black line. The sensor readings are as following figures (2 3 4 5 6 7 8):



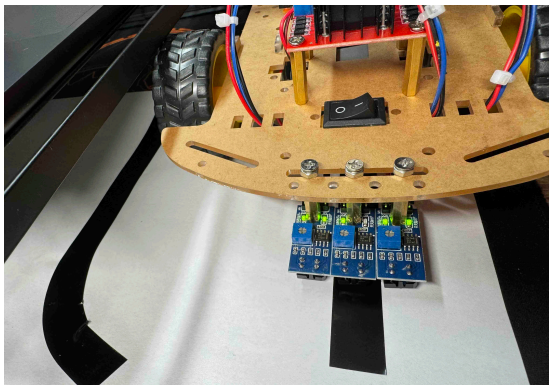
```
LEFT: 58        MIDDLE: 66      RIGHT: 39
LEFT: 58        MIDDLE: 66      RIGHT: 39
LEFT: 58        MIDDLE: 66      RIGHT: 39
LEFT: 58        MIDDLE: 66      RIGHT: 39
LEFT: 58        MIDDLE: 66      RIGHT: 39
LEFT: 58        MIDDLE: 66      RIGHT: 39
LEFT: 58        MIDDLE: 66      RIGHT: 39
LEFT: 58        MIDDLE: 66      RIGHT: 39
LEFT: 58        MIDDLE: 66      RIGHT: 39
LEFT: 58        MIDDLE: 66      RIGHT: 39
LEFT: 58        MIDDLE: 66      RIGHT: 39
LEFT: 58        MIDDLE: 66      RIGHT: 39
LEFT: 58        MIDDLE: 66      RIGHT: 39
LEFT: 58        MIDDLE: 66      RIGHT: 39
LEFT: 58        MIDDLE: 66      RIGHT: 39
LEFT: 58        MIDDLE: 66      RIGHT: 39
```

Figure 2: When the sensor reads 000



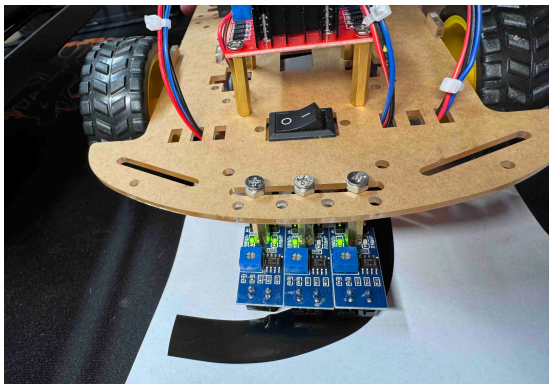
```
LEFT: 58        MIDDLE: 68      RIGHT: 774
LEFT: 58        MIDDLE: 68      RIGHT: 775
LEFT: 58        MIDDLE: 69      RIGHT: 775
LEFT: 58        MIDDLE: 69      RIGHT: 776
LEFT: 58        MIDDLE: 68      RIGHT: 773
LEFT: 59        MIDDLE: 68      RIGHT: 772
LEFT: 59        MIDDLE: 69      RIGHT: 774
LEFT: 59        MIDDLE: 68      RIGHT: 776
LEFT: 58        MIDDLE: 69      RIGHT: 772
LEFT: 58        MIDDLE: 69      RIGHT: 776
LEFT: 58        MIDDLE: 69      RIGHT: 781
LEFT: 62        MIDDLE: 83      RIGHT: 1023
LEFT: 61        MIDDLE: 81      RIGHT: 1023
LEFT: 62        MIDDLE: 82      RIGHT: 1023
LEFT: 62        MIDDLE: 82      RIGHT: 1023
LEFT: 62        MIDDLE: 82      RIGHT: 1023
LEFT: 62        MIDDLE: 83      RIGHT: 1023
LEFT: 62        MIDDLE: 83      RIGHT: 1023
```

Figure 3: When the sensor reads 001



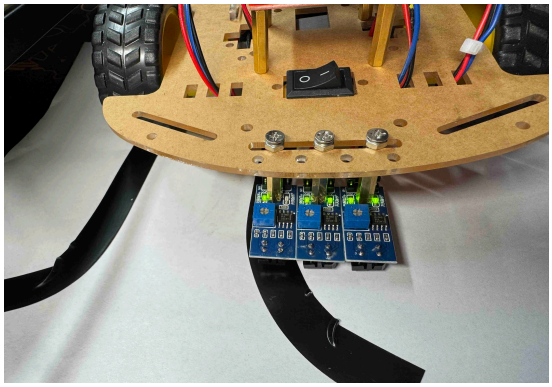
```
LEFT: 49        MIDDLE: 1023    RIGHT: 58
LEFT: 46        MIDDLE: 1023    RIGHT: 52
LEFT: 36        MIDDLE: 1023    RIGHT: 52
LEFT: 55        MIDDLE: 1023    RIGHT: 47
LEFT: 35        MIDDLE: 918     RIGHT: 50
LEFT: 39        MIDDLE: 1023    RIGHT: 65
LEFT: 55        MIDDLE: 1023    RIGHT: 48
LEFT: 55        MIDDLE: 1023    RIGHT: 50
LEFT: 55        MIDDLE: 1023    RIGHT: 48
LEFT: 56        MIDDLE: 1023    RIGHT: 48
LEFT: 56        MIDDLE: 1023    RIGHT: 48
LEFT: 56        MIDDLE: 1023    RIGHT: 48
LEFT: 56        MIDDLE: 1023    RIGHT: 48
LEFT: 55        MIDDLE: 1023    RIGHT: 48
LEFT: 55        MIDDLE: 1023    RIGHT: 48
```

Figure 4: When the sensor reads 010



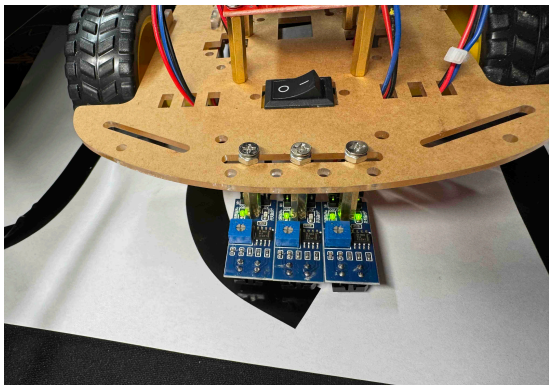
```
LEFT: 50        MIDDLE: 1023    RIGHT: 1023
LEFT: 50        MIDDLE: 1023    RIGHT: 1023
LEFT: 50        MIDDLE: 1023    RIGHT: 1023
LEFT: 50        MIDDLE: 1023    RIGHT: 1023
LEFT: 50        MIDDLE: 1023    RIGHT: 1023
LEFT: 50        MIDDLE: 1023    RIGHT: 1023
LEFT: 45        MIDDLE: 971     RIGHT: 1023
LEFT: 44        MIDDLE: 1023    RIGHT: 1023
LEFT: 43        MIDDLE: 1023    RIGHT: 1023
LEFT: 44        MIDDLE: 1023    RIGHT: 1023
LEFT: 45        MIDDLE: 1023    RIGHT: 1023
LEFT: 43        MIDDLE: 227     RIGHT: 1023
LEFT: 42        MIDDLE: 91      RIGHT: 1023
LEFT: 42        MIDDLE: 207     RIGHT: 1023
LEFT: 42        MIDDLE: 1023    RIGHT: 1023
LEFT: 43        MIDDLE: 1023    RIGHT: 1023
LEFT: 47        MIDDLE: 1023    RIGHT: 1023
LEFT: 47        MIDDLE: 1023    RIGHT: 1023
LEFT: 47        MIDDLE: 1023    RIGHT: 1023
```

Figure 5: When the sensor reads 011



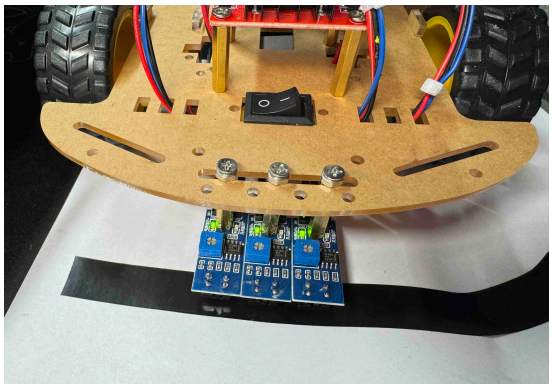
```
LEFT: 1023      MIDDLE: 74      RIGHT: 57
LEFT: 1023      MIDDLE: 76      RIGHT: 56
LEFT: 1023      MIDDLE: 78      RIGHT: 57
LEFT: 1023      MIDDLE: 74      RIGHT: 57
LEFT: 1023      MIDDLE: 75      RIGHT: 56
LEFT: 1023      MIDDLE: 74      RIGHT: 56
LEFT: 1023      MIDDLE: 74      RIGHT: 56
LEFT: 1023      MIDDLE: 74      RIGHT: 56
LEFT: 1023      MIDDLE: 73      RIGHT: 57
LEFT: 1023      MIDDLE: 74      RIGHT: 57
LEFT: 1023      MIDDLE: 73      RIGHT: 57
LEFT: 1023      MIDDLE: 74      RIGHT: 57
LEFT: 1023      MIDDLE: 74      RIGHT: 56
LEFT: 1023      MIDDLE: 74      RIGHT: 56
LEFT: 1023      MIDDLE: 74      RIGHT: 57
LEFT: 1023      MIDDLE: 74      RIGHT: 57
LEFT: 1023      MIDDLE: 74      RIGHT: 57
LEFT: 1023      MIDDLE: 74      RIGHT: 57
LEFT: 1023      MIDDLE: 73      RIGHT: 57
LEFT: 1023      MIDDLE: 73      RIGHT: 56
```

Figure 6: When the sensor reads 100



```
LEFT: 901       MIDDLE: 992     RIGHT: 43
LEFT: 930       MIDDLE: 1017    RIGHT: 43
LEFT: 950       MIDDLE: 1023    RIGHT: 43
LEFT: 965       MIDDLE: 1023    RIGHT: 43
LEFT: 970       MIDDLE: 1023    RIGHT: 43
LEFT: 977       MIDDLE: 1023    RIGHT: 44
LEFT: 992       MIDDLE: 1023    RIGHT: 44
LEFT: 996       MIDDLE: 1023    RIGHT: 44
LEFT: 1001      MIDDLE: 1023    RIGHT: 44
LEFT: 1004      MIDDLE: 1023    RIGHT: 44
LEFT: 1006      MIDDLE: 1023    RIGHT: 44
LEFT: 1019      MIDDLE: 1023    RIGHT: 44
LEFT: 1023      MIDDLE: 1023    RIGHT: 44
LEFT: 1023      MIDDLE: 1023    RIGHT: 44
LEFT: 1023      MIDDLE: 1023    RIGHT: 44
LEFT: 1023      MIDDLE: 1023    RIGHT: 44
```

Figure 7: When the sensor reads 110



```
LEFT: 1023      MIDDLE: 1023    RIGHT: 1023
LEFT: 1023      MIDDLE: 1023    RIGHT: 1023
LEFT: 1023      MIDDLE: 1023    RIGHT: 1023
LEFT: 1023      MIDDLE: 1023    RIGHT: 1023
LEFT: 1023      MIDDLE: 1023    RIGHT: 1023
LEFT: 1023      MIDDLE: 1023    RIGHT: 1023
LEFT: 1023      MIDDLE: 1023    RIGHT: 1023
LEFT: 1023      MIDDLE: 1023    RIGHT: 1023
LEFT: 1023      MIDDLE: 1023    RIGHT: 1023
LEFT: 1023      MIDDLE: 1023    RIGHT: 1023
LEFT: 1023      MIDDLE: 1023    RIGHT: 1023
LEFT: 1023      MIDDLE: 1023    RIGHT: 1023
LEFT: 1023      MIDDLE: 1023    RIGHT: 1023
LEFT: 1023      MIDDLE: 1023    RIGHT: 1023
LEFT: 1023      MIDDLE: 1023    RIGHT: 1023
LEFT: 1023      MIDDLE: 1023    RIGHT: 1023
```

Figure 8: When the sensor reads 111

4 Results

4.1 Experiment 1

Through experiments and observations from images, it can be seen that when the infrared sensor is placed on a white surface, the infrared light is effectively reflected after being emitted and can be received. At this time, the module outputs a low voltage level, and the output indicator light is illuminated. However, when the infrared sensor is placed on a black surface, black objects almost do not reflect infrared light. Therefore, when the line-following module detects a black line, it outputs a high voltage level, and the output indicator light is turned off. Additionally, when the line-following module is positioned too high above the ground, its detection effect is similar to that when encountering a black line. It will also output a high voltage level, and the indicator light will be turned off.
We tested three sets of values at different horizontal heights of the sensor above the ground and calculated their average values as the final results 1.
The relationship between the detection values and the height was ultimately represented by a graph 9.

Altitude/mm	**Data**			**Average**
5	33	34	33	33.3
10	37	38	38	37.7
15	42	43	42	42.3
20	48	49	48	48.3
25	59	54	58	57.0
30	65	70	86	73.7
35	294	275	315	294.7
40	560	465	510	511.7
45	765	680	648	697.7
50	895	800	846	847.0

Table 1: Data and Average Values

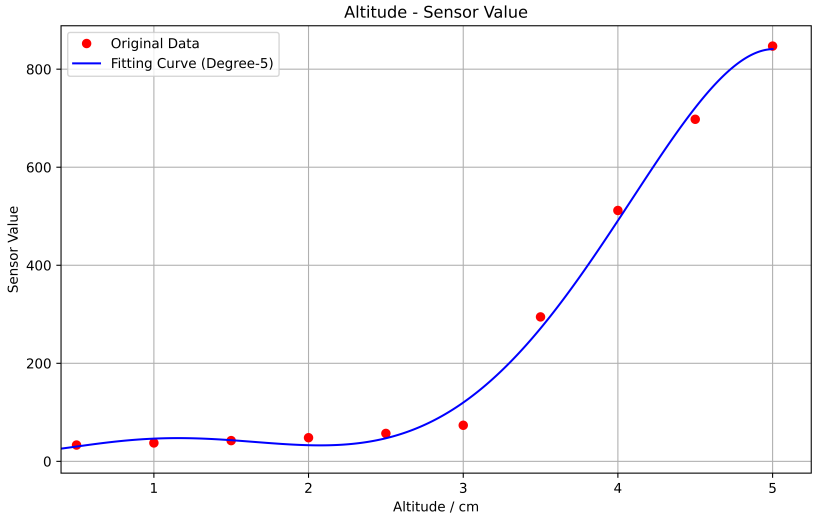


Figure 9: Altitude - Sensor Value Graph

Based on the data above, it can be seen that when the sensor is 3 cm above the ground, the sensor values undergo a sudden change. Therefore, in order to ensure stability, the sensor must be positioned no more than 3 cm above the ground.

4.2 Experiment 2

Based on the aforementioned experiment, we first recorded the sensor readings. Subsequently, we established a recognition criterion: when the sensor reading is less than 100, it is marked as 0̈; and when the reading approaches 1023, it is marked as 1̈. Here, 0̈indicates that the sensor has detected a white surface, while 1̈indicates that it has detected a black line. Under these conditions, the sensor readings under different scenarios are shown in the figures (10 11 12 13 14 15 16) below:

```
LEFT: 0; MIDDLE: 0; LIGHT: 0
LEFT: 0; MIDDLE: 0; LIGHT: 0
LEFT: 0; MIDDLE: 0; LIGHT: 0
LEFT: 0; MIDDLE: 0; LIGHT: 0
LEFT: 0; MIDDLE: 0; LIGHT: 0
LEFT: 0; MIDDLE: 0; LIGHT: 0
LEFT: 0; MIDDLE: 0; LIGHT: 0
LEFT: 0; MIDDLE: 0; LIGHT: 0
LEFT: 0; MIDDLE: 0; LIGHT: 0
LEFT: 0; MIDDLE: 0; LIGHT: 0
```

Figure 10: When the sensor reads 000

```
LEFT: 0; MIDDLE: 0; LIGHT: 1
LEFT: 0; MIDDLE: 0; LIGHT: 1
LEFT: 0; MIDDLE: 0; LIGHT: 1
LEFT: 0; MIDDLE: 0; LIGHT: 1
LEFT: 0; MIDDLE: 0; LIGHT: 1
LEFT: 0; MIDDLE: 0; LIGHT: 1
LEFT: 0; MIDDLE: 0; LIGHT: 1
LEFT: 0; MIDDLE: 0; LIGHT: 1
LEFT: 0; MIDDLE: 0; LIGHT: 1
LEFT: 0; MIDDLE: 0; LIGHT: 1
```

Figure 11: When the sensor reads 001

```
LEFT: 0; MIDDLE: 1; LIGHT: 0
LEFT: 0; MIDDLE: 1; LIGHT: 0
LEFT: 0; MIDDLE: 1; LIGHT: 0
LEFT: 0; MIDDLE: 1; LIGHT: 0
LEFT: 0; MIDDLE: 1; LIGHT: 0
LEFT: 0; MIDDLE: 1; LIGHT: 0
LEFT: 0; MIDDLE: 1; LIGHT: 0
LEFT: 0; MIDDLE: 1; LIGHT: 0
LEFT: 0; MIDDLE: 1; LIGHT: 0
LEFT: 0; MIDDLE: 1; LIGHT: 0
LEFT: 0; MIDDLE: 1; LIGHT: 0
LEFT: 0; MIDDLE: 1; LIGHT: 0
```

Figure 12: When the sensor reads 010

```
LEFT: 0; MIDDLE: 1; LIGHT: 1
LEFT: 0; MIDDLE: 1; LIGHT: 1
LEFT: 0; MIDDLE: 1; LIGHT: 1
LEFT: 0; MIDDLE: 1; LIGHT: 1
LEFT: 0; MIDDLE: 1; LIGHT: 1
LEFT: 0; MIDDLE: 1; LIGHT: 1
LEFT: 0; MIDDLE: 1; LIGHT: 1
LEFT: 0; MIDDLE: 1; LIGHT: 1
LEFT: 0; MIDDLE: 1; LIGHT: 1
LEFT: 0; MIDDLE: 1; LIGHT: 1
LEFT: 0; MIDDLE: 1; LIGHT: 1
LEFT: 0; MIDDLE: 1; LIGHT: 1
```

Figure 13: When the sensor reads 011

```
LEFT: 1; MIDDLE: 0; LIGHT: 0
LEFT: 1; MIDDLE: 0; LIGHT: 0
LEFT: 1; MIDDLE: 0; LIGHT: 0
LEFT: 1; MIDDLE: 0; LIGHT: 0
LEFT: 1; MIDDLE: 0; LIGHT: 0
LEFT: 1; MIDDLE: 0; LIGHT: 0
LEFT: 1; MIDDLE: 0; LIGHT: 0
LEFT: 1; MIDDLE: 0; LIGHT: 0
LEFT: 1; MIDDLE: 0; LIGHT: 0
LEFT: 1; MIDDLE: 0; LIGHT: 0
LEFT: 1; MIDDLE: 0; LIGHT: 0
```

Figure 14: When the sensor reads 100

```
LEFT: 1; MIDDLE: 1; LIGHT: 0
LEFT: 1; MIDDLE: 1; LIGHT: 0
LEFT: 1; MIDDLE: 1; LIGHT: 0
LEFT: 1; MIDDLE: 1; LIGHT: 0
LEFT: 1; MIDDLE: 1; LIGHT: 0
LEFT: 1; MIDDLE: 1; LIGHT: 0
LEFT: 1; MIDDLE: 1; LIGHT: 0
LEFT: 1; MIDDLE: 1; LIGHT: 0
LEFT: 1; MIDDLE: 1; LIGHT: 0
LEFT: 1; MIDDLE: 1; LIGHT: 0
LEFT: 1; MIDDLE: 1; LIGHT: 0
LEFT: 1; MIDDLE: 1; LIGHT: 0
```

Figure 15: When the sensor reads 110

```
LEFT: 1; MIDDLE: 1; LIGHT: 1
LEFT: 1; MIDDLE: 1; LIGHT: 1
LEFT: 1; MIDDLE: 1; LIGHT: 1
LEFT: 1; MIDDLE: 1; LIGHT: 1
LEFT: 1; MIDDLE: 1; LIGHT: 1
LEFT: 1; MIDDLE: 1; LIGHT: 1
LEFT: 1; MIDDLE: 1; LIGHT: 1
LEFT: 1; MIDDLE: 1; LIGHT: 1
LEFT: 1; MIDDLE: 1; LIGHT: 1
LEFT: 1; MIDDLE: 1; LIGHT: 1
LEFT: 1; MIDDLE: 1; LIGHT: 1
LEFT: 1; MIDDLE: 1; LIGHT: 1
```

Figure 16: When the sensor reads 111

5 Conclusions

Conclusion and Summary

In this experiment, we accomplished the following tasks:

1. By conducting multiple measurements of the height between the infrared sensor and the ground and the corresponding changes in readings, we ensured the accuracy of the

readings and the reliability of the data, and ultimately plotted a variation curve. With the help of this curve, we were able to determine the optimal height range for installing the infrared sensor, which provided a scientific basis for the sensor installation position in subsequent experiments and laid a solid foundation.

2. By analyzing the differences in readings of the infrared sensor on the white ground and the black line, we established the identification criteria for different reading ranges corresponding to the ground and the line to be followed. This achievement provided strong technical support for the subsequent line-following tasks, ensuring the accuracy and reliability of the system.

Limitations of the experiment

1. Due to the limitations of our experimental conditions, we could only measure the distance between the sensor and the ground manually, which may have caused some deviations in the sensor readings.

2. Our experiment only considered the scenario where the ground is white under the same lighting conditions. This might lead to interference for the robot car when encountering different colored surfaces during the actual line-following process.

Appendix

Data Process

There are two parts of the appendix, the first of which is a **Python** code for data processing.

```python
import numpy as np
import mlx.core as mx
import matplotlib.pyplot as plt

def graphGenerate():
    altitude = [
        0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0
    ]

    sensor_stanley = [
        33, 37, 42, 48, 59, 65, 294, 560, 765, 895
    ]

    sensor_yilin = [
        34, 38, 43, 49, 54, 70, 275, 465, 680, 800
    ]

    sensor_chloe = [
        33, 38, 42, 48, 58, 86, 315, 510, 648, 846
    ]

    sensor_average = [
        (s + y + c) / 3 for s, y, c in zip(sensor_stanley,
           sensor_yilin, sensor_chloe)
    ]

    sensorValue = mx.array(sensor_average)
    altitudeValue = mx.array(altitude)

    poly_degree = 5
    poly_coeffs = np.polyfit(altitudeValue, sensorValue, poly_degree)
    poly_func = np.poly1d(poly_coeffs)

    altitude_dense = mx.linspace(0, 5, 600)
    sensor_dense = poly_func(altitude_dense)

    plt.figure(figsize=(10, 6))
    plt.plot(altitudeValue, sensorValue, 'ro', label='Original Data')
    plt.plot(altitude_dense, sensor_dense, 'b-', label=f"Fitting
       Curve (Degree-{poly_degree})")

    plt.xlim(left=0.4)

    plt.xlabel("Altitude / cm")
    plt.ylabel("Sensor Value")
    plt.title('Altitude - Sensor Value')
```

```
    plt.grid(True)
    plt.legend()

    plt.savefig("./Photos/lab4_graph.png", dpi = 1200)
    plt.show()

if __name__ == '__main__':
    graphGenerate()
```

Origin Code

The second part is the code of **Arduino IDE** for parameters gaining.

```
void setup() {
  // put your setup code here, to run once:
  Serial.begin(9600);
}

void loop() {
  int sensorValue_LEFT = analogRead(A4);
  delay(2);
  int sensorValue_MIDDLE = analogRead(A2);
  delay(2);
  int sensorValue_RIGHT = analogRead(A0);
  delay(2);

  Serial.print("LEFT: ");
  Serial.print(printStatus(sensorValue_LEFT));
  Serial.print("; MIDDLE: ");
  Serial.print(printStatus(sensorValue_MIDDLE));
  Serial.print("; LIGHT: ");
  Serial.println(printStatus(sensorValue_RIGHT));

  delay(1000);
}

int printStatus(int value) {
  int ans = 0;
  if(value > 800) {
    ans = 1;
  }
  return ans;
}
```